Password Storage

1. Why is storing passwords in plain text considered insecure? Because anyone who accesses the database can read the passwords directly.

2. What is a password hash? A fixed-length value generated from a password using a hashing algorithm.

3. How does hashing differ from encryption? Hashing is designed to be one-way, while encryption can be reversed using the correct key.

4. What is the purpose of a salt? A salt is a random value added before hashing to make identical passwords produce different hashes.

5. Name two modern password hashing algorithms. Argon2id and scrypt.

Password Auditing

1. What is the purpose of John the Ripper? To audit password strength and identify weak passwords.

2. What is the purpose of Hashcat? To test password guesses against hashes for password auditing and recovery.

3. Do password auditing tools decrypt passwords? No. They generally test password guesses and compare the resulting hashes with stored hashes.

4. Why do organizations perform password audits? To identify weak or compromised passwords and improve security.

5. Why should password auditing only be performed with authorization? To protect privacy, comply with laws, and prevent unauthorized access.

Password Security

1. What makes a password strong? A password that is long, unique, unpredictable, and difficult to guess.

2. Why is password reuse dangerous? If one account is compromised, attackers may use the same password to access other accounts.

3. How does a password manager improve security? It securely stores passwords and generates strong, unique passwords for different accounts.

4. What is Multi-Factor Authentication (MFA)? A security method that requires two or more different types of verification to access an account.

5. Why do organizations implement account lockout policies? To limit repeated failed login attempts and help prevent brute-force attacks.

Defense Strategies

1. What is a password policy? A set of rules that defines how users should create and manage passwords.

2. Why is defense in depth important? It provides multiple layers of security so that if one control fails, others can still protect the system.

3. How does user awareness contribute to password security? It helps users recognize phishing, avoid password reuse, and protect their credentials.

4. Why is monitoring login activity important? It helps detect suspicious login attempts and potential unauthorized access.

5. How do multiple security controls work together to protect user accounts? Password policies, secure hashing, MFA, account lockout, and login monitoring work together to reduce the risk of unauthorized access.